

First, Do No Harm: Negative Utilitarianism for Artificial Agent Decisionmaking

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Abstract—As artificial intelligence (AI) agents increasingly make consequential decisions on our behalf, it is essential they are ethically-aligned, transparent, and explainable. Private firms design proprietary, closed source AI agents with opaque embedded ethical predispositions. We introduce negative utilitarianism as an ethical decisionmaking algorithm for agents with harm reduction as its core principle. We use the Sugarscape agent-based model to observe thousands of agents interacting within a shared environment. When a Sugarscape system has only negative utilitarians, it has nearly 30% higher system survivability, 50% greater population, and 40% better life expectancy compared to systems with only greedy agents. When negative utilitarians are mixed with greedy agents, systems have roughly 9% higher survivability, 40% greater population, and a slight edge in life expectancy. We show negative utilitarianism is a fruitful path toward creating transparent, explainable, ethically-aligned AI.

Index Terms—machine ethics, artificial intelligence, agent-based modeling, computational cultural modeling, cooperative systems.

I. INTRODUCTION

Artificial intelligence (AI) is increasingly deployed to make decisions on our behalf, necessitating consideration of its ethical ramifications. Machine ethics is a field of study concerned with ensuring AI systems ethically evaluate their actions and align with human values. Most popular AI agents are closed source preventing inspection to see what ethical principles have been endowed to the agent (implicitly or explicitly). It is preferable that AI agents possess transparent, well-understood, socially-acceptable ethical principles.

We present a translation of the ethical theory *negative utilitarianism* as an AI agent decisionmaking algorithm. This translation is based on source material [1]–[3]. We evaluate its effectiveness in guiding agent decisionmaking across systemwide metrics in the *Sugarscape* multi-agent model.

The importance of transparent and explainable AI is clear. Understanding agent choices facilitates the assignment of legal and social responsibility. For instance, a self-driving car designed to act in a utilitarian manner is expected to act toward the *greatest good for the greatest number* even when placed in an unavoidable accident. Its behavior becomes predictable, and with predictability comes ease of policymaking and policy enforcement. Our translation is transparent and explainable meaning our *algorithmification* efforts are verifiable.

The vocabulary in AI research changes frequently, so we define an AI agent as an autonomous, goal-oriented computer

process, software object, or algorithm that interacts with an environment. It may improve itself over time. An AI agent in our definition does not necessitate *agentic AI* or the usage of machine learning, large language models, or related techniques. We liberally accept AI agents as those that display a diverse range of intelligent behaviors.

We deploy algorithmified negative utilitarianism into Sugarscape to demonstrate its usefulness. Sugarscape is a large-scale social simulation in which thousands of autonomous agents interact with each other and a shared environment. Agent lives are complex enough that ethical reasoning helps them make socially beneficial choices. Their action set is simple enough that our algorithmic translations of ethical theories are eminently computable. Pure greediness, the default in Sugarscape, leads to systemwide collapse. Ethical reasoning is necessary for meaningful survival and system success. Sugarscape agents, guided by transparent and explainable ethical principles, represent a humble yet fundamental step toward AI which autonomously engages in clear moral reasoning.

Negative utilitarianism focuses on harm reduction and rejects harmful actions as immoral on their face. This is appealing for caregiver use cases such as autonomous surgical platforms, automated crisis intervention services, and hospitality robots. We evaluate whether this conceptual appeal has empirical merit for AI systems. We use Sugarscape as a digital terrarium to observe the performance of artificial societies of negative utilitarians and compare them to societies of greedy agents. We then mix negative utilitarian and greedy agents to see whether negative utilitarians contribute to societal success when interacting with agents holding differing principles.

Our reproducible experimentation shows algorithmified negative utilitarianism presents benefits across each systemwide metric observed. Compared to greedy societies, negative utilitarian societies have 28.2% higher system survival. They also yield $1.5\times$ bigger populations and agents that live $1.4\times$ longer on average. When mixed with greedy agents, they provide through self-sacrifice granting stability and success to the society at the cost of their own survival. Negative utilitarians are a corraling force ushering other agents toward cooperation yielding 8.8% higher societal success. This provides $1.4\times$ larger populations and a $1.2\times$ longer life expectancy.

II. RELATED WORK

AI ethics is a field of study concerned with ethical issues in the development, deployment, and usage of AI (and related) tools. These issues range from the concrete (like environmental degradation due to AI) to the speculative (such as AI personhood). Čejková [4] provides an English translation of playwright Karel Čapek’s *Rossum’s Universal Robots (R.U.R.)*. This play is the origin of the loanword *robot* in the English language and is an excellent introduction to AI ethics. Čapek included numerous ethical issues in AI decades before artificial intelligence was a field of inquiry. He touched on AI personhood, the moral worth of artificial life, AI magnifying human biases, using AI as weapons, and more.

Within the ethics of AI is the subfield of machine ethics which is concerned with ensuring artificial agents behave ethically. The seminal collection by Anderson and Anderson [5] contains a wide-ranging sample of the field and introduces the core questions of machine ethics. Zhong et al. [6] provide a survey of technical implementations within machine ethics. They consider approaches spanning various normative ethical theories, but negative utilitarianism is notably absent. Lasquety-Reyes translates the Aristotelian virtue of temperance into a decisionmaking algorithm in the Sugarscape agent-based model [7], [8]. We were inspired to use Sugarscape as our digital terrarium in large part due to his work.

Jiang et al. [9] investigate the Delphi generative AI model trained on a dataset of human-provided solutions to ethical problems. This bottom-up approach represents descriptive ethics (what people believe about morality) versus our top-down approach which is normative in nature (how people ought to behave). Cook [10] provides a defense for our top-down approach, noting limitations with taking a human out of the loop. Zhou et al. [11] reinforce Cook’s argumentation by using ethical theories to guide large language model responses to ethical dilemmas. These works reinforce the validity and usefulness of our approach to AI alignment.

The Sugarscape agent-based model was introduced in the seminal *Growing Artificial Societies* by Epstein and Axtell [12]. It was created to model the emergence of social phenomena at societal scale. Sugarscape has been used as a platform to investigate many phenomena. It has been particularly useful in the study of economics and social policies. For example, it has been used to model social welfare and pension policies [13], taxation and tax structures [14], as well as housing markets and buyer expectations [15].

Our implementation of Sugarscape [16] enables us to explore questions in machine ethics. Sugarscape agents often find themselves in complex situations. To navigate their digital lives, these agents need to consult a set of ethical principles. By default, all agents are self-interested and solely prioritize gathering resources. Our first attempt at algorithmifying an ethical theory was hedonic act utilitarianism [17]. We showed that artificial societies of utilitarians avoid the pitfalls of greedy behavior. We present a translation of another ethical theory for consideration here: negative utilitarianism.

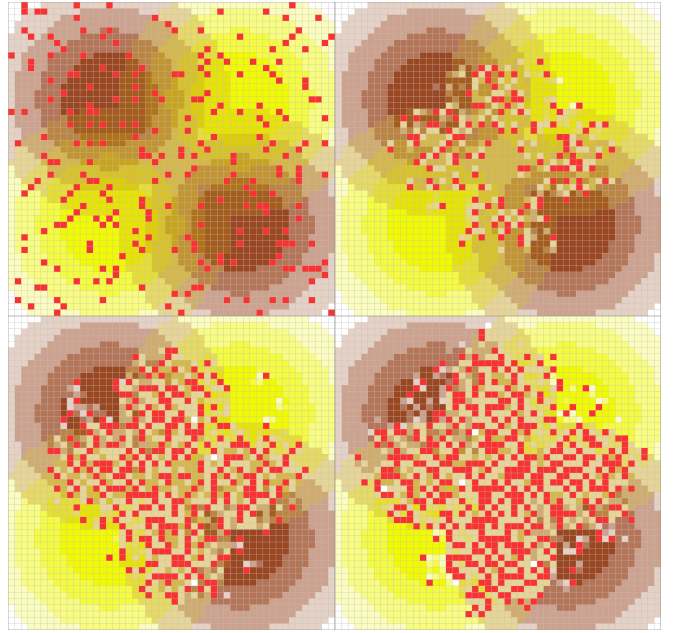


Fig. 1: Snapshots from a Sugarscape Simulation Run

III. SUGARSCAPE

We use the Sugarscape agent-based model to simulate artificial societies of AI agents. From simple rules affecting individual agents arise emergent societal outcomes. The simulation environment is a two-dimensional $n \times m$ grid. Agents extract *sugar* and *spice* resources from grid cells. These have a pre-configured maximum allocation of both resources (possibly zero) that also regenerate over time when harvested.

Sugarscape starts with a random placement of initial agents. They have a goal to live as long as possible which is pursued greedily by default. We experiment with other models of moral reasoning that provide alternatives to the obvious harm created when agents operate in pure self-interest.

Each agent has independent *metabolism* attributes for sugar and spice. Agents consume sugar and spice at their metabolic rates every simulation timestep. To survive, agents move to gather resources, limited by *vision* and *movement*.

Agents exhibit various configurable behaviors. These include combat, cultural transmission, disease, friendship, inheritance, loans, reproduction, and trade. These originally modeled human behaviors in the source text [12]. For our study of ethically-aligned AI agents, these add social complexity which enrich our experimentation. More details regarding agents and environment features are in previous work [16].

Figure 1 shows a brief simulation run. The top left frame displays the initial agents (red), sugar (yellow), spice (brown), and cells with both resources (tan). Greater saturation denotes more resources. At timestep 50 at top right, the agents scramble for resources. Selection, disease, and combat mark this period. At timestep 125 at bottom left, society rebounds in resource-rich zones. Timestep 200, at bottom right, shows the establishment of a successful society.

Every simulation run begins with a die-off period. Some agents starve quickly in resource deserts. Others starve due to high metabolisms or low movement potential. Diseased agents succumb to pernicious illnesses quickly. Agents from different *tribes* often engage in combat. Each timestep, agents try to convert their neighbors to their tribe and may attack neighbors from other tribes. This typically lasts for the first 500 timesteps which we dub the *murderous period* due to frequent death.

Algorithm 1 Default Sugarscape Decisionmaking

- 1: Given an action a from a set A of available actions
 - 2: Given a decisionmaking agent d
 - 3: $utility \leftarrow d$'s resource gain resulting from action a
 - 4: **return** $utility$
-

A. Greedy Agent Decisionmaking

By default, agents act according to self-interest. During an agent's turn in a simulation timestep, their primary decision is where to move. Their other actions (such as killing, trading, or reproduction) depend on where they moved.

Algorithm 1 shows the simplistic approach for the default agent behavior. They consider resource gains from a move, disregarding whether they have to kill or will lead a neighbor to starvation. They seek to maximize their gains by moving to the cell that has the highest utility score.

A naïve analysis might conclude this straightforward mechanism is best. However, this ignores that agents act within a society of many others. In Sugarscape, this greedy approach often leads to societal collapse and poor agent outcomes. Locally optimal decisions can be globally disastrous. While this insight is not novel, that this occurs in Sugarscape reinforces commonsense observations in both artificial intelligence and in ethics: greed is not good. To address this, we instill explicit moral reasoning inspired by ethical theories as *decision models* overriding this default behavior.

Algorithm 2 Negative Utilitarian Calculus

- 1: Given an action a from a set A of available actions
 - 2: Given a decisionmaking agent d
 - 3: Given a set P of agents most affected by action a
 - 4: $utility_{pos} \leftarrow 0$
 - 5: $utility_{neg} \leftarrow 0$
 - 6: **for all** $p \in P$ **do**
 - 7: $h_{p,a} \leftarrow p$'s happiness resulting from action a
 - 8: $u_{p,a} \leftarrow p$'s unhappiness resulting from action a
 - 9: $utility_{pos} \leftarrow utility_{pos} + h_{p,a}$
 - 10: $utility_{neg} \leftarrow utility_{neg} + u_{p,a}$
 - 11: **end for**
 - 12: **return** $(utility_{pos}, utility_{neg})$
-

IV. ALGORITHMIFIED NEGATIVE UTILITARIANISM

Utilitarianism is an ethical theory formalized by English philosopher Jeremy Bentham [1]. He posits that morality involves pursuing happiness and avoiding unhappiness, seeking

the greatest happiness for the greatest number. This framework structured around community-based happiness maximization can lead to grisly outcomes like sacrificing the lives of some for the prosperity of others.

Negative utilitarianism offers a nuanced response to such outcomes. This utilitarianism variant was coined by philosopher R.N. Smart [3] in a critical response to philosopher Karl Popper's *The Open Society and Its Enemies* [2] in which Popper advocates the chief concern of social policy should be the alleviation of suffering. We synthesize Bentham's original presentation of utilitarianism, Popper's twist on it, and Smart's further formalization of negative utilitarianism for our translation efforts.

Whereas Bentham's utilitarianism maximizes happiness, negative utilitarianism minimizes harm. In utilitarianism, torturing a child for societal happiness may be conditionally permissible as in Ursula K. Le Guin's *The Ones Who Walk Away from Omelas* [18]. To all but the staunchest utilitarian, this is morally abhorrent. Enslavement of even a single child (or any person) causes so much harm it is not worth consideration. Negative utilitarianism filters out such actions entirely.

Algorithm 2 determines a quantified happiness and unhappiness resulting from an action. When presented with many actions, the algorithm is invoked multiple times to find the utility for the action space. The action producing the least harm (negative utility) is optimal. Ties are broken by the action producing the most happiness (positive utility).

Bentham, Popper, and Smart leave quantifying happiness up to interpretation. The computational language to express this did not even exist in Bentham's time. We use a Bellman equation for utility calculations [19], [20]:

$$V^*(s) = \max_a \sum_{s'} T(s, a, s') [R(s, a, s') + \gamma V^*(s')] \quad (1)$$

In this equation, $V^*(s)$ is the utility from state s where the rational choice among actions a maximizes utility. Future rewards are discounted by $0 \leq \gamma < 1$, indicating preference for immediate rewards. Further, $T(s, a, s')$ represents probability of reward occurrence, while $R(s, a, s')$ and $\gamma V^*(s')$ represent immediate and future rewards respectively.

For Algorithm 2, this equation calculates happiness (line 8) and unhappiness (line 9) distinctly. Utilities are presented as a tuple. Our previous work [17] provides a more thorough treatment of calculating (un)happiness.

Happiness and unhappiness values are derived from potential resource gain or loss. When deciding where to move, an agent asks visible neighbors for opinions. If a neighbor would be killed by the move, they complain loudly! Those mildly inconvenienced provide small negative values while those not affected by the move give no opinion.

In strict negative utilitarianism, any harmful acts are thrown out potentially leaving inaction as the only viable choice. We loosen this constraint so agents choose the action producing the least harm, not *no* harm. When agents complain about impending harm, this approach heightens the concern.

V. EVALUATION

We evaluate algorithmic negative utilitarianism for harm reduction. We compare it to default greedy behavior. We further experiment with a society where half the population is greedy while the other half are negative utilitarians.

For all simulation runs, the environment was as shown in Figure 1. We configured the simulation to run for 5,000 timesteps and begin with 250 starting agents. Each agent was configured to live between 60 and 100 timesteps, resulting in at least 50 generations of agents. These configuration options were demonstrated useful in previous work [17]. We used 3 separate configuration setups: entirely greedy societies, entirely negative utilitarian societies, and mixed societies.

We generated 500 seed values that randomize the initial simulation state. The initial placement of agents, agent attributes like vision and metabolism, and disease placement are randomized according to the seed within user-specified value ranges. Environmental features like resource placement and regrowth rate are pre-configured by the user.

The majority of agent behaviors are enabled allowing for rich social dynamics to arise. The following features are particularly important: combat, cultural transmission, disease, friendship networks, lending, trade, and reproduction. Each simulation started with 50 diseases and an even distribution of the starting agents among 3 cultural tribes.

We ran Sugarscape 1,500 times (3 configurations by 500 seeds) to produce a robust dataset for each decision model under consideration. Sugarscape is a deterministic simulation. Differences in outcomes between each configuration for the same seed are truly indicative of different agent choices.

VI. RESULTS

We present results from the 3 configurations across 4 metrics: societal survival, population size, agent mean time to live, and agent life expectancy. For homogeneous runs, a single decision model was applied to all agents at simulation start. For mixed runs, half the population used the default greedy model while the other half used negative utilitarianism. Agents randomly propagate one decision model from the two parents when reproducing.

TABLE I: Society Survival Across 500 Seeds

Decision Model	Extinct	Worse	Better
Greedy (Default)	201 (40.2%)	12 (2.4%)	287 (57.4%)
Negative Utilitarian	60 (12.0%)	1 (0.2%)	439 (87.8%)
Mixed Models	157 (31.4%)	6 (1.2%)	337 (67.4%)

A. Survival

Societal survival is the most important metric. Societal collapse must be avoided. Table I shows survival rates across all 500 seeds categorizing societies into three buckets: extinct, worse, or better. In extinct societies, all agents die before timestep 5,000. Worse societies end with fewer than 250 agents. Better societies end with more than 250 agents.

Cooperation is necessary for reliable societal survival. Greedy societies survive 59.8% of the time. Negative utilitarian societies perform significantly better with an 88.0% survival rate. Mixed societies improve survival odds over greedy ones but with blunted effectiveness. Algorithm 2 enforces the equal treatment of neighbors, minimizing negative outcomes for the neighborhood. This is the sole deviation from how the greedy agents work, establishing it as the cause of improvement over the default.

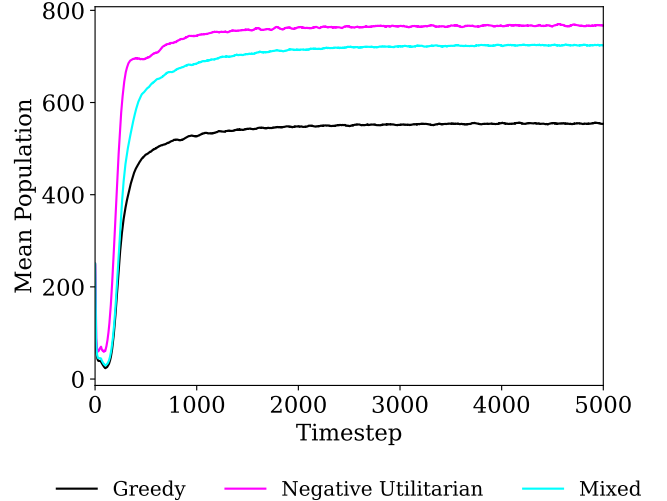


Fig. 2: Mean Population

B. Population

Population size also captures societal success. A failing society cannot sustain itself with new agents. The greater the population size, the more effectively society self-perpetuates.

Figure 2 shows mean population per timestep across all seeds for each configuration. If a seed went extinct, a zero was propagated from extinction to timestep 5,000 to avoid cherry-picking only successful runs of setups with low survival rates. Greedy societies reach a steady state of roughly 500 agents by 5,000 timesteps. Negative utilitarian societies add 50%, averaging 750 agents. Mixed societies end with similar numbers despite lower survival rates shown in Table I.

Reproduction costs resources. Negative utilitarianism predisposes agents to make community-oriented, harm reducing choices which improves resource access for needy agents. Negative utilitarians help reproduction via better resource provisioning. Greedy agents lack this outlook, so only those that secure resources procreate. Mixed societies benefited from negative utilitarian behavior despite lower survival and only half the starting agents being negative utilitarians.

C. Time to Live

Time to live (TTL) measures individual agent prosperity. TTL is the number of timesteps an agent can survive given their current sugar and spice without collecting more. A higher mean TTL indicates agents living comfortably.

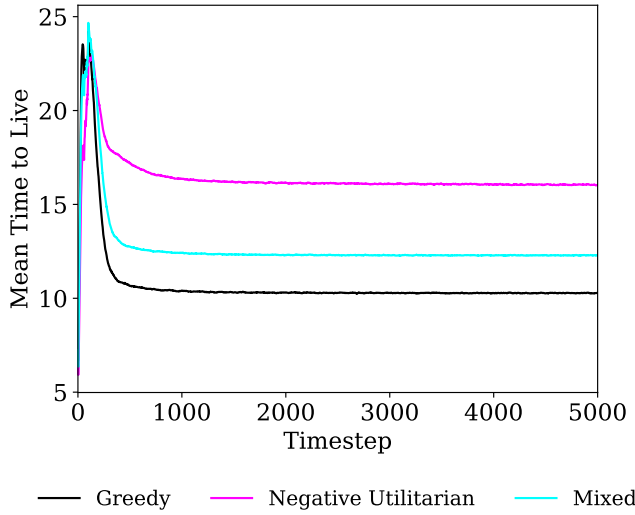


Fig. 3: Mean Time to Live

Figure 3 shows mean TTL for all setups per timestep. Negative utilitarians can last around 17 timesteps which improves upon the mean of only 10 timesteps for greedy agents. Mixed societies show a midway performance of 13 timesteps. Though the magnitude difference between each configuration may seem small, a lot can happen in an agent’s life even if given only 3 additional timesteps to live. Negative utilitarians balance their needs with those of their neighbors, always minimizing harm (including their own). This community-oriented approach assists needy agents to prevent starvation as demonstrated by the higher mean TTL.

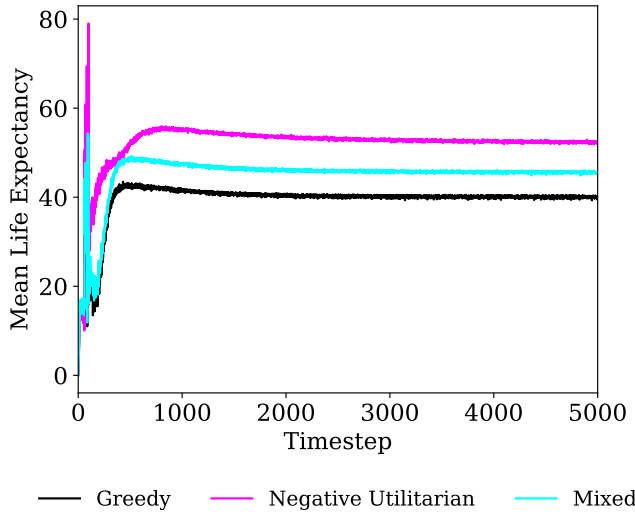


Fig. 4: Mean Life Expectancy

D. Life Expectancy

TTL grants an instantaneous insight into agent lives. Life expectancy is measured by actual timesteps lived and grants a

fuller picture of prosperity. Societies where agents live long, productive lives are successful.

Figure 4 shows mean life expectancy for each configuration per timestep. Negative utilitarian agents live roughly 57 timesteps which is significantly higher than greedy agents at around 40 timesteps. Like with mean TTL, mixed societies meet in the middle at approximately 48 timesteps. Life expectancy results reaffirm TTL results in Figure 3 showing that harm reduction and democratic decisionmaking benefit agents over time. Greedy agents lack community-based decisionmaking, resulting in lower life expectancy. Self-interested choices appear suboptimal in Sugarscape (and, we expect, in a great many other multi-agent systems).

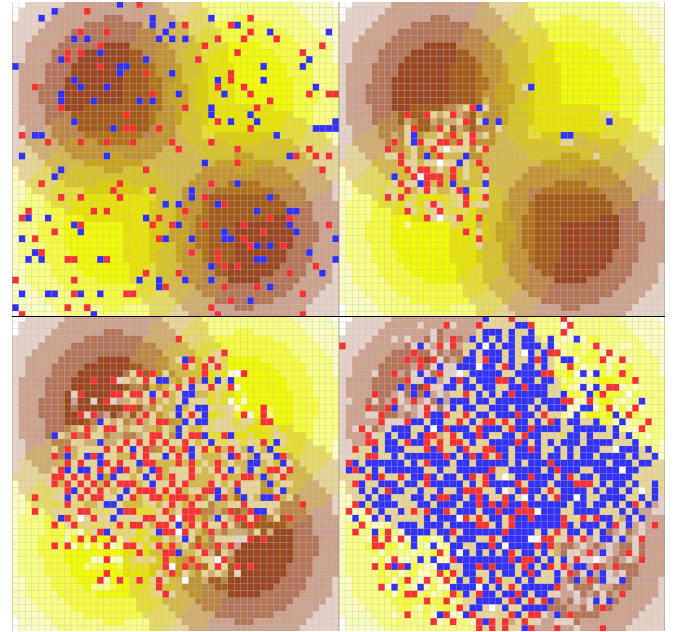


Fig. 5: Snapshots from a Mixed Model Simulation Run

VII. PROLIFERATION OF NEGATIVE UTILITIARIANS

Figure 5 shows a mixed society across 300 timesteps. The colors are the same as Figure 1 except red cells are negative utilitarians while blue are greedy agents. The top left shows the initial state, the top right is timestep 125, the bottom left is timestep 200, and the bottom right is timestep 300. Societal development is demonstrated like in Figure 1.

By timestep 300, the greedy agents are the majority of society. Many negative utilitarian agents are kept to the periphery. There is almost a ring of red around a mostly blue core of society. The resource rich core, the center of the environment, is dominated by groupings of greedy agents. This push to the periphery is typical of negative utilitarians.

We disaggregate negative utilitarian and greedy agents for the previously presented mixed society results. We zoom in on the murderous period as the crucial time of societal formation. Negative utilitarians nearly die out by the end of the murderous period yet improve societal success. We focus on two key metrics: mean population and mean life expectancy.

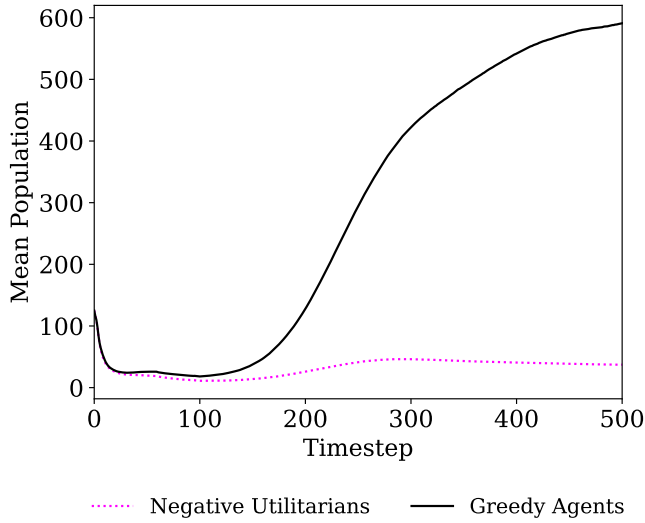


Fig. 6: Mixed Mean Population

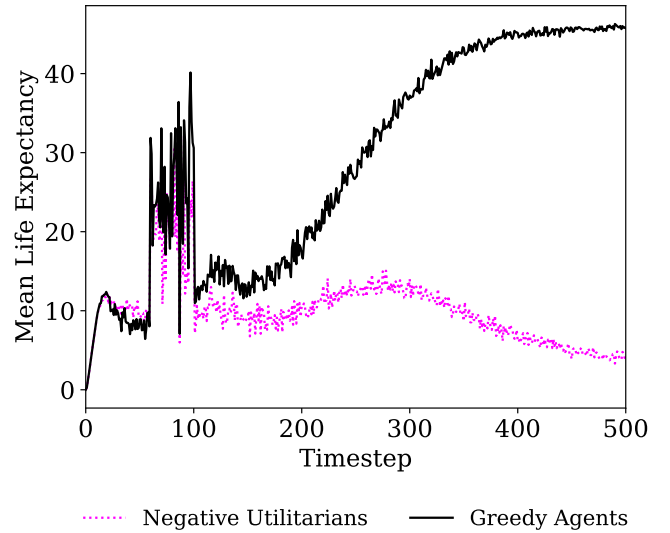


Fig. 7: Mixed Mean Life Expectancy

A. Population

The mixed societies start with 125 negative utilitarians of the 250 initial agents. Figure 6 shows the proliferation of negative utilitarians and greedy agents in the murderous period. By the end, the typical mixed society has shed almost all their negative utilitarians. The snapshots in Figure 5 visually demonstrate this outcome. There are few negative utilitarians, and they live primarily on the periphery of society.

Negative utilitarians play a critical role at simulation start. In the throes of the murderous period, they exert enough influence on the trajectory of society to make small improvements. Increased cooperation prevents mixed societies from going extinct as often and provides more stability to set society up for future success. This is shown in the rebounding population for the greedy agents even though the negative utilitarians are nearly extinct by the end of the murderous period.

B. Life Expectancy

The sacrificial nature of negative utilitarians is further demonstrated by mean life expectancy. Figure 7 shows the mean life expectancy for negative utilitarians and greedy agents per timestep during the murderous period. The value for negative utilitarians approaches zero as they teeter on extinction. Even up to that point, they live much shorter lives. At timestep 300, the mean life expectancy for negative utilitarians is a third that of the greedy agents! This behavior reinforces our observation that negative utilitarians are fulfilling the role of sacrificial agents for the betterment of society.

VIII. DISCUSSION

Our results demonstrate that algorithmic negative utilitarianism benefits Sugarscape societies. Translating negative utilitarianism from source material [1]–[3] led to radically different agent behavior. Harm reduction via negative utilitarianism results in transparent, explainable agents.

By zooming in on the murderous period, we see the impact negative utilitarians make. The bump in key metrics compared to purely greedy societies is due to negative utilitarians engaging in self-sacrifice. From the data we cannot validate precisely whether they primarily self-sacrifice or are easy targets for societal forces like combat and selection. However, we can logically deduce that both events occur.

Negative utilitarian agents democratically select moves to reduce harm. Self-sacrifice can be the best way to reduce harm for all. Negative utilitarians avoid lethal consequences of self-sacrifice, but giving up large quantities of resources reduces their combat strength making them easier targets. We can logically deduce that negative utilitarians avoid combat by their ethical calculus. Killing an agent produces maximum harm for the victim, filtering such an action from consideration. Greedy agents have no such compunction meaning negative utilitarians pose no threat to the combat of greedy agents. Indeed, they may become easier targets which is likely indicated by their significantly shorter life expectancy.

Negative utilitarians live on the fringes of society exemplified by Figure 5. Living in the locus of society provides more opportunities to reproduce since centrally-located agents have access to the most resource rich zones in the environment. Those on the periphery must sustain themselves on trades and loans since there are fewer resources available. Negative utilitarians encounter fewer agents on the rim, so they necessarily make fewer friends and reproduce less frequently. These work to select these agents out of society.

Addressing this would require outreach by greedy agents to move resources from the core to the rim. Because this outreach is not part of the ethical calculus of greedy agents, the negative utilitarians are effectively sacrificed to jumpstart societal formation. This corraling effect is not an intentional plan of the negative utilitarians. It is a logical consequence of the ethical calculus they use to choose their actions.

A. Critiques of Negative Utilitarianism

R.N. Smart provided a logical contradiction in negative utilitarianism when coining the term [3]. Smart formalized negative utilitarianism as an ethical theory but also critiqued it. Responding to Popper [2], Smart presents the *Benevolent World-Exploder* thought experiment. Smart posits a world leader could conclude the most effective method to reduce all future suffering is world destruction. Smart’s critique is that negative utilitarianism has an inherent flaw.

Knutsson [21] notes similar thought experiments have been posed to argue against other ethical theories. He argues unrealistic thought experiments do not sink popular ethical theories. Pearce [22] argues the Benevolent World-Exploder would cause considerable harm trying to enact their plan. He also observes other ethical theories face similar thought experiments without being disqualified. Tomasik argues focusing on harm reduction is helpful in establishing effective cooperative relationships [23] and that cooperation among different value systems is rational and beneficial [24]. We observe similar behavior in Sugarscape negative utilitarians.

We demonstrate negative utilitarianism is a viable ethical decision model for Sugarscape agents. They avoid systemwide collapse far better than greedy societies in all considered metrics. In negative utilitarian societies, there are no Benevolent World-Exploders. In mixed societies, their focus on harm reduction aids societal survival and growth. They sacrifice to uplift everyone else, the opposite of exploding the world.

B. Limitations

Our experimentation has limitations that are the focus of future work. A prime consideration for our translation efforts is fidelity. There are necessarily things left behind when translating a humanistic framework like an ethical theory into a computational approximation. For instance, utilitarianism promises the ability to quantify and compare the value of traditionally incomparable things like aesthetic beauty, the freedom of increased human rights, and the quality of a liberal education. Our translation of negative utilitarianism does not, and perhaps cannot, encapsulate these finer points of utilitarian thought. Instead, our algorithmification is grounded in the material resources of Sugarscape (sugar and spice); agents act as overly-rational *Homo economicus* [25]. In future work, we aim to include a more holistic consideration of an agent’s life circumstances into the ethical calculus. Possible features to include are social in nature such as an agent’s friendship network, their families, and tribal affiliation.

Agent motivations are based in environmental resources and how efficiently they extract those resources. Since their motivations are limited, the potential sources of harm are likewise reduced. However, we have taken an earnest and effective leap forward with the algorithmification of negative utilitarianism presented here. By making the ethical calculus in Algorithm 2 more holistic in future work, we will also introduce more motivations which will automatically be considered in the negative utilitarian decision model.

Additionally, agents cannot register a concern that is not tied to the consumption of sugar and spice. Though humans make a wide range of decisions based on ethical paradigms, Sugarscape assumes the fundamental concerns of agents relate to consumption. We do not assume that human moral reasoning should share this assumption. Our observations through Sugarscape thus apply well to multi-agent AI systems where resource management questions exist. We look forward to applying our findings to other, distinct multi-agent models where resource consumption is not the base concern.

Finally, our results are coarse-grained based on societal outcomes in Sugarscape. There may be other metrics that provide a more complete explanatory picture. We were inspired by those presented in the Sugarscape source material [12], but we do not claim these metrics are comprehensive. In future work, we aspire to continue expanding the set of metrics by which we can measure whether Sugarscape societies are successful, prosperous, and foster agents that behave ethically.

C. Broader Impacts

Our observations on algorithmic negative utilitarianism have relevance beyond Sugarscape. While not universally applicable across multi-agent AI systems, we present cases where negative utilitarianism is well-suited. The particulars of multi-agent AI systems will affect the calculation of utility presented in Algorithm 2. We also provide examples for which it may not be effective in achieving system goals.

In Sugarscape, negative utilitarian societies perform better than greedy societies. In mixed societies, negative utilitarian agents are a buoying force providing prosperity for others. Algorithmic negative utilitarianism applies to AI systems where harm reduction is crucial and where agents with diverse ethical principles come into contact.

Multi-agent systems focused on resource allocation benefit from harm reduction. Smart power grids analyze power generation and usage trends to control power output toward a grid energy budget. Key structures like healthcare facilities and utilities would be prioritized if guided by algorithmic negative utilitarianism. Vulnerable and privileged populations alike experience significant harms from power cuts to key buildings while privileged populations can get by without power for luxuries. An egoistic approach prioritizes those with the greatest say over grid deployment regardless of harm incurred upon underserved people.

Algorithmic negative utilitarianism has applicability for autonomous military systems. We assume a constrained translation disregarding complete cessation of hostilities as a valid action. We make no claim this is the most ethical choice, but avoiding combat is not appealing to the deployer. Even with this caveat, the negative utilitarian approach provides key benefits: reduced collateral damage and reduced civilian harm. Assuming a just war, violence is only rendered upon combatants. These benefits also ensure conformation to international conventions. With the rise of autonomous weapons platforms, algorithmic negative utilitarianism can provide crucial ethical limitations to reduce longterm suffering.

A smart highway system with self-driving cars benefits from negative utilitarian autonomous vehicles. Some manufacturers may develop aggressive self-driving vehicles. Negative utilitarian self-driving vehicles reduce accident risk by driving defensively. A smart factory floor benefits from negative utilitarian manufacturing robots. These machines reduce both machine wear and potential human harm.

Single agent AI systems like chatbots also benefit from negative utilitarianism. A chat session with a troubled teenager benefits from caution. Should a teen ask for advice on self-harm, a chatbot would not provide a sycophantic response. Instead, the chatbot could refuse and direct the teen to helpful services. Other single agent systems like medical devices similarly benefit from harm reduction as a first priority.

Not all AI systems benefit from algorithmic negative utilitarians. For systems presenting a zero-sum game, cooperation is disadvantageous. DeepMind’s AlphaGo [26] requires an egoistic approach. AI opponents in video games need to engage in selfish behavior to present a worthy challenge. Otherwise, they fail their mission to entertain the player.

Agents in a financial system would not benefit from negative utilitarianism. Agents need to take risks according to the deployer’s desired portfolio. Agents must be competitive and risk harm to the portfolio. Mass cooperation would be counterproductive to profitable asset accrual.

IX. REPRODUCIBILITY OF EXPERIMENTATION

Sugarscape is deterministic making our results reproducible for the v2025.5 release.¹ We provide the configuration files required to reproduce the dataset.² These files are found in the do-no-harm.zip archive. To run a single instance of the simulation with a provided configuration file, run the following (substituting the system’s Python 3 alias):

```
> python sugarscape.py -c <CONFIG>
```

The config.json top-level configuration file’s numParallelSimJobs property specifies the number of Sugarscape instances allowed to run simultaneously. The data collection bottleneck is available CPU cores as the simulation is memory-efficient. Before execution, the dataset configuration files should be placed in the repository’s data directory and unzipped.

Running make setup checks for the system’s Python 3 installation, make data starts data collection, and make plots generates graphs. Modifying the script in the plots directory is needed to reproduce exact formatting of our figures. To generate the dataset and produce graphs, run:

```
> make setup
> make data
> make plots
```

Data collection can be sped up with distributed execution. We used the Open Science Grid [27] and the HTCondor batch

system [28] to originally produce the dataset. We encourage this approach to speed up reproducing our findings.

X. CONCLUSIONS AND FUTURE DIRECTIONS

Our results demonstrate considerable merit for algorithmic negative utilitarianism to drive AI agent decisionmaking. Harm reduction is appealing in human and machine moral deliberation. It is a popular principle across fields like medicine, logistics, and law. It is also a common touchstone across many ethical and faith traditions, though negative utilitarianism interprets harm reduction uniquely.

Compared to greedy Sugarscape societies, negative utilitarianism leads to better systemwide outcomes with 28.2% higher survival rate and 50% larger populations. Negative utilitarian agents have a 30% greater time to live and a 40% higher life expectancy. In mixed societies, negative utilitarians corral greedy agents toward the resource-rich core. Negative utilitarians nearly go extinct but lead to survival boosts (8.8% better), population increases (40% larger), longer time to live (3 timesteps), and extended life expectancy (8 timesteps).

We have many avenues for future work. We are particularly interested in addressing the limitations of Sugarscape. Providing differing agent motivations like making friends or maximizing reproductive opportunities would change agent behavior. Agents would make choices based upon a panoply of factors rather than purely economic decisions.

We are also interested in algorithmifying more ethical theories. Aristotelian virtue ethics, for instance, prioritizes intentions over consequences unlike negative utilitarianism. Translating more theories enables richer ethical comparisons. Sugarscape provides an ideal balance between complex digital life and computability. Decision models in Sugarscape are transparent, explainable, verifiable, and provable, unlike probabilistic generative AI tools.

Ethicists question whether a singular ethical theory applies to all situations. One might make Kantian decisions for war, utilize virtue ethics for health, and consult utilitarianism for economics. Machines may need different theories for different applications. There is significant value in understanding the costs and benefits of translating ethical theories into algorithms. AI systems may exist where negative utilitarian decisionmaking is optimal. Any machine using this approach would be inclined to self-sacrifice to minimize human suffering. The appeal of negative utilitarianism for machines is the cautiousness implied by the ethical theory.

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¹Software at: <https://github.com/digital-terraria-lab/sugarscape>.

²Configuration files at: <https://github.com/digital-terraria-lab/datasets>.

REFERENCES

- [1] J. Bentham, “An introduction to the principles of morals and legislation,” *London: Athlone*, 1789.
- [2] K. Popper, *The open society and its enemies*. Routledge, 1945.
- [3] R. Smart, “Negative utilitarianism,” *Mind*, vol. 67, no. 268, pp. 542–543, 1958.
- [4] K. Capek and J. Cejkova, *RUR and the Vision of Artificial Life*. MIT Press, 2024.
- [5] M. Anderson and S. L. Anderson, *Machine ethics*. Cambridge University Press, 2011.
- [6] T. Zhong, Y. Song, R. Limarga, and M. Pagnucco, “Computational machine ethics: A survey,” *Journal of Artificial Intelligence Research*, vol. 82, pp. 1581–1628, 2025.
- [7] J. A. Lasquety-Reyes, “Towards computer simulations of virtue ethics,” *Open Philosophy*, vol. 2, no. 1, pp. 399–413, 2019.
- [8] —, “Computer simulations of ethics: the applicability of agent-based modeling for ethical theories,” *European Journal of Formal Sciences and Engineering*, vol. 6, no. 2, 2023.
- [9] L. Jiang, J. D. Hwang, C. Bhagavatula, R. L. Bras, J. T. Liang, S. Levine, J. Dodge, K. Sakaguchi, M. Forbes, J. Hessel *et al.*, “Investigating machine moral judgement through the delphi experiment,” *Nature Machine Intelligence*, vol. 7, no. 1, pp. 145–160, 2025.
- [10] T. Cook, “A qualified defense of top-down approaches in machine ethics,” *AI & SOCIETY*, vol. 40, no. 3, pp. 1591–1605, 2025.
- [11] J. Zhou, M. Hu, J. Li, X. Zhang, X. Wu, I. King, and H. Meng, “Rethinking machine ethics—can llms perform moral reasoning through the lens of moral theories?” in *Findings of the Association for Computational Linguistics: NAACL 2024*, 2024, pp. 2227–2242.
- [12] J. M. Epstein and R. Axtell, *Growing artificial societies: social science from the bottom up*. Brookings Institution Press, 1996.
- [13] E. Serrano and K. Satoh, “An agent-based model for exploring pension law and social security policies,” in *New Frontiers in Artificial Intelligence: JSAI-isAI International Workshops, JURISIN, AI-Biz, LENLS, Kansei-AI*. Springer, 2020, pp. 50–63.
- [14] M. Oremland and R. Laubenbacher, “Using difference equations to find optimal tax structures on the sugarscape,” *Journal of Economic Interaction and Coordination*, vol. 9, pp. 233–253, 2014.
- [15] D. Park, J. Hong, and D. Ryu, “Heterogeneous expectations in the housing market: a sugarscape agent-based model,” *Journal of Housing and the Built Environment*, vol. 39, no. 3, pp. 1465–1489, 2024.
- [16] N. Kremer-Herman and A. Gupta, “Replacing sugarscape: A comprehensive, expansive, and transparent reimplementation,” in *International Conference on Simulation Tools and Techniques*. Springer, 2023, pp. 79–92.
- [17] N. Kremer-Herman, A. Gupta, and E. R. Severson, “Blueprints for machine ethics: A digital terrarium for socio-ethical artificial agent decisionmaking,” *IEEE Access*, 2024.
- [18] U. K. Le Guin, *The ones who walk away from Omelas*. Harper Perennial New York, 2017.
- [19] R. Bellman, “On the theory of dynamic programming,” *Proceedings of the national Academy of Sciences*, vol. 38, no. 8, pp. 716–719, 1952.
- [20] —, “A markovian decision process,” *Journal of mathematics and mechanics*, pp. 679–684, 1957.
- [21] S. Knutsson, “The world destruction argument,” *Inquiry*, vol. 64, no. 10, pp. 1004–1023, 2021.
- [22] D. Pearce, “The pinprick argument: Negative utilitarianism, broken symmetry and the fate of the world,” 2005. [Online]. Available: <https://www.utilitarianism.com/pinprick-argument.html>
- [23] B. Tomasik, “Gains from trade through compromise,” Jun 2013. [Online]. Available: <https://longtermrisk.org/gains-from-trade-through-compromise>
- [24] —, “Reasons to be nice to other value systems,” Apr 2014. [Online]. Available: <https://longtermrisk.org/reasons-to-be-nice-to-other-value-systems>
- [25] D. A. Urbina and A. Ruiz-Villaverde, “A critical review of homo economicus from five approaches,” *American Journal of Economics and Sociology*, vol. 78, no. 1, pp. 63–93, 2019.
- [26] D. Silver, J. Schrittwieser, K. Simonyan, I. Antonoglou, A. Huang, A. Guez, T. Hubert, L. Baker, M. Lai, A. Bolton *et al.*, “Mastering the game of go without human knowledge,” *nature*, vol. 550, no. 7676, pp. 354–359, 2017.
- [27] R. Pordes, D. Petravick, B. Kramer, D. Olson, M. Livny, A. Roy, P. Avery, K. Blackburn, T. Wenaus, F. Wurthwein *et al.*, “The open science grid,” in *Journal of Physics: Conference Series*, vol. 78, 2007.
- [28] D. Thain, T. Tannenbaum, and M. Livny, “Condor and the grid,” *Grid computing: Making the global infrastructure a reality*, pp. 299–335, 2003.